

Solution to The MidSem Exam

Paper 1

$$y_i = \alpha + \beta x_i + \varepsilon_i, \quad i=1, \dots, n, \quad E(\varepsilon_i) = 0, \quad \text{var}(\varepsilon_i) = \sigma^2$$

$\varepsilon_1, \dots, \varepsilon_n$ are iid

(a) Least-squares estimators are obtained by minimizing

$$\frac{1}{n} \sum_{i=1}^n (y_i - a - b x_i)^2 \quad \text{w.r.t. } (a, b).$$

This was done in the class.

$$\hat{\beta}_{LSE} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{\sum_{i=1}^n (x_i - \bar{x}) y_i}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

$$\hat{\alpha}_{LSE} = \bar{y} - \hat{\beta}_{LSE} \bar{x}$$

$$\text{var}(\hat{\beta}_{LSE}) = \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}, \quad \text{var}(\hat{\alpha}_{LSE}) = \sigma^2 \left\{ \frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \right\}$$

These were derived in the class.

(b) Gauss-Markov Theorem: Under the given setup

$$(y_i = \alpha + \beta x_i + \varepsilon_i, \quad E(\varepsilon_i) = 0, \quad \text{var}(\varepsilon_i) = \sigma^2, \quad \varepsilon_i \text{'s are uncorrelated})$$

$(\hat{\alpha}_{LSE}, \hat{\beta}_{LSE})$ ^(as a pair) is the best linear unbiased estimator of (α, β) ^(as a pair).

Consequently, $\hat{\alpha}_{LSE}$ is the BLUE of α

$\hat{\beta}_{LSE}$ is the BLUE of β .

(c) $(\hat{\alpha}_\lambda, \hat{\beta}_\lambda)$ are obtained by minimizing

$$L(a, b) = \frac{1}{n} \sum_{i=1}^n (y_i - a - bx_i)^2 + \lambda b^2$$

w.r.t. a & b .

These can be obtained by differentiating $L(a, b)$ w.r.t. a & b and setting them to 0. Doing that would lead to normal equations:

$$\begin{aligned} a + b\bar{x} &= \bar{y} \\ a\bar{x} + b(\bar{x}^2 + \lambda) &= \bar{x}\bar{y} \end{aligned} \quad \left| \begin{aligned} \bar{x}^2 &= \frac{1}{n} \sum_{i=1}^n x_i^2 \\ \bar{x}\bar{y} &= \frac{1}{n} \sum_{i=1}^n x_i y_i \end{aligned} \right.$$

two equations in two variables

Solving these gives

$$\begin{aligned} b &= \frac{\bar{x}\bar{y} - \bar{x}\bar{y}}{\bar{x}^2 - \bar{x}^2 + \lambda} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2 + n\lambda} \\ &= \frac{\sum_{i=1}^n (x_i - \bar{x}) y_i}{\sum_{i=1}^n (x_i - \bar{x})^2 + n\lambda} \end{aligned}$$

$$a = \bar{y} - b\bar{x}$$

$$\therefore \hat{\beta}_\lambda = \frac{\sum_{i=1}^n (x_i - \bar{x}) y_i}{\sum_{i=1}^n (x_i - \bar{x})^2 + n\lambda}, \quad \hat{\alpha}_\lambda = \bar{y} - \hat{\beta}_\lambda \bar{x}$$

It can be verified that the Hessian is p.d.

$$(d) \hat{\beta}_\lambda = \sum_{i=1}^n w_{i\lambda} y_i, \quad w_{i\lambda} = \frac{x_i - \bar{x}}{\sum_{i=1}^n (x_i - \bar{x})^2 + n\lambda}$$

$$\therefore \text{Var}(\hat{\beta}_\lambda) = \sum_{i=1}^n w_{i\lambda}^2 \text{Var}(y_i) \quad \left[\begin{array}{l} y_i \text{'s are indep} \\ \text{Var}(y_i) = \sigma^2 \end{array} \right]$$

$$= \sigma^2 \sum_{i=1}^n w_{i\lambda}^2$$

$$= \sigma^2 \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{\left\{ \sum_{i=1}^n (x_i - \bar{x})^2 + n\lambda \right\}^2}$$

$$\hat{\alpha}_\lambda = \bar{y} - \hat{\beta}_\lambda \bar{x} = \sum_{i=1}^n \left(\frac{1}{n} - \bar{x} w_{i\lambda} \right) y_i$$

$$\therefore \text{Var}(\hat{\alpha}_\lambda) = \sigma^2 \sum_{i=1}^n \left(\frac{1}{n} - \bar{x} w_{i\lambda} \right)^2$$

$$= \sigma^2 \left\{ \frac{1}{n} + \bar{x}^2 \sum_{i=1}^n w_{i\lambda}^2 - \cancel{\frac{2\bar{x}}{n} \sum_{i=1}^n w_{i\lambda}} \right\}$$

$$= \sigma^2 \left[\frac{1}{n} + \frac{\bar{x}^2 \sum_{i=1}^n (x_i - \bar{x})^2}{\left\{ \sum_{i=1}^n (x_i - \bar{x})^2 + n\lambda \right\}^2} \right]$$

$$(e) \text{Var}(\hat{\alpha}_{\text{LSE}}) = \sigma^2 \left[\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \right]$$

$$= \sigma^2 \left[\frac{1}{n} + \frac{\bar{x}^2 \sum_{i=1}^n (x_i - \bar{x})^2}{\left\{ \sum_{i=1}^n (x_i - \bar{x})^2 \right\}^2} \right]$$

$$\text{Var}(\hat{\alpha}_\lambda) = \sigma^2 \left[\frac{1}{n} + \frac{\bar{x}^2 \sum_{i=1}^n (x_i - \bar{x})^2}{\left\{ \sum_{i=1}^n (x_i - \bar{x})^2 + n\lambda \right\}^2} \right]$$

$$\therefore \lambda > 0, \quad \sum_{i=1}^n (x_i - \bar{x})^2 + n\lambda > \sum_{i=1}^n (x_i - \bar{x})^2 \dots \textcircled{*}$$

The only difference between the two variances are the denominators $\bigcirc$. The inequality $\textcircled{*}$ gives $\text{Var}(\hat{\alpha}_\lambda) < \text{Var}(\hat{\alpha}_{\text{LSE}})$

$$\text{Var}(\hat{\beta}_{\text{LSE}}) = \sigma^2 \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{\left\{ \sum_{i=1}^n (x_i - \bar{x})^2 \right\}^2}$$

$$\text{Var}(\hat{\beta}_\lambda) = \sigma^2 \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{\left\{ \sum_{i=1}^n (x_i - \bar{x})^2 + n\lambda \right\}^2}$$

The difference is only in the denominator. Again, (*) gives $\text{Var}(\hat{\beta}_\lambda) < \text{Var}(\hat{\beta}_{\text{LSE}})$.

Although we get estimators with lower variances than the LSE's, this does not violate the Gauss-Markov Theorem.

$$\begin{aligned} \text{Here, } E(\hat{\beta}_\lambda) &= \sum_{i=1}^n w_{i\lambda} (d + \beta x_i) \\ &= \cancel{\alpha \sum_{i=1}^n w_{i\lambda}} + \beta \sum_{i=1}^n w_{i\lambda} x_i \\ &= \beta \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{\sum_{i=1}^n (x_i - \bar{x})^2 + n\lambda} \neq \beta \quad (\because \lambda > 0) \end{aligned}$$

$\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{\sum_{i=1}^n (x_i - \bar{x})^2 + n\lambda}$

Similarly,

$$\begin{aligned} E(\hat{\alpha}_\lambda) &= E(\bar{y}) - \bar{x} E(\hat{\beta}_\lambda) \\ &= d + \beta \bar{x} - c\beta \bar{x}, \quad c = \\ &\neq d \quad (\because c \neq 1) \end{aligned}$$

Although $\hat{\alpha}_\lambda$ & $\hat{\beta}_\lambda$ are linear, they are not unbiased. Gauss-Markov Theorem is about linear unbiased estimators.

Paper 2

$$y_i = \alpha + \beta x_i + \varepsilon_i, \quad i=1, \dots, n, \quad \varepsilon_1, \dots, \varepsilon_n \stackrel{i.i.d.}{\sim} N(0, \sigma^2).$$

To test $H_0: \beta = 0$ vs. $H_a: \beta \neq 0$.

$$(a) \begin{pmatrix} \hat{\alpha}_{LSE} \\ \hat{\beta}_{LSE} \end{pmatrix} \sim N_2 \left(\begin{pmatrix} \alpha \\ \beta \end{pmatrix}, \sigma^2 \begin{pmatrix} n & \sum_{i=1}^n x_i \\ \sum_{i=1}^n x_i & \sum_{i=1}^n x_i^2 \end{pmatrix}^{-1} \right)$$

$$\Rightarrow \hat{\beta}_{LSE} \sim N \left(\beta, \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \right) \quad \left[\begin{array}{l} \text{variance already} \\ \text{calculated in Paper 1} \end{array} \right]$$

$$\frac{SSE}{\sigma^2} \sim \chi_{n-2}^2, \quad SSE = \sum_{i=1}^n (y_i - \hat{\alpha}_{LSE} - \hat{\beta}_{LSE} x_i)^2$$

$\hat{\beta}_{LSE}$ & SSE are independent.

F test statistic is

$$F = \frac{\hat{\beta}_{LSE}^2 \sum_{i=1}^n (x_i - \bar{x})^2}{SSE / (n-2)} \sim F_{1, n-2} \quad \text{under } H_0$$

Reject H_0 at level α if $F > F_{\alpha; 1, n-2}$

↑ upper α -quantile.

$$(b) \quad SSE = \sum_{i=1}^n (y_i - \hat{\alpha} - \hat{\beta} x_i)^2, \quad \hat{\alpha} = \bar{y} - \hat{\beta} \bar{x}, \quad \hat{\beta} = \frac{S_{xy}}{S_{xx}}$$
$$S_{xy} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$
$$S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2$$

$$\Rightarrow SSE = \sum_{i=1}^n (y_i - \bar{y} - \hat{\beta}(x_i - \bar{x}))^2$$

$$= S_{yy} + \hat{\beta}^2 S_{xx} - 2\hat{\beta} S_{xy}$$

$$\hat{\beta} S_{xy} = \frac{S_{xy}^2}{S_{xx}} = \hat{\beta}^2 S_{xx}$$

$$\Rightarrow SSE = S_{yy} - \hat{\beta}^2 S_{xx}$$

Now,

$$\begin{aligned} F &= (n-2) \frac{\hat{\beta}^2 S_{xx}}{SSE} = (n-2) \frac{\hat{\beta}^2 S_{xx}}{S_{yy} - \hat{\beta}^2 S_{xx}} \\ &= (n-2) \frac{\hat{\beta}^2 S_{xx} / S_{yy}}{1 - \hat{\beta}^2 S_{xx} / S_{yy}}, \quad \hat{\beta}^2 \frac{S_{xx}}{S_{yy}} = \frac{S_{xy}^2}{S_{xx} S_{yy}} = r^2 \\ &= (n-2) \frac{r^2}{1-r^2} \end{aligned}$$

(c) Given : $n = 50$, $\sum x_i = 20$, $\sum y_i = 30$, $\sum x_i^2 = 40$,
 $\sum y_i^2 = 36$, $\sum x_i y_i = 10$

$$\therefore \bar{x} = \frac{2}{5}, \quad \bar{y} = \frac{3}{5}$$

$$S_{xx} = \sum x_i^2 - n \bar{x}^2 = 40 - 50 \times \frac{4}{25} = 32$$

$$S_{yy} = \sum y_i^2 - n \bar{y}^2 = 36 - 50 \times \frac{9}{25} = 18$$

$$S_{xy} = \sum x_i y_i - n \bar{x} \bar{y} = 10 - 50 \times \frac{2}{5} \times \frac{3}{5} = -2$$

$$r^2 = \frac{S_{xy}^2}{S_{xx} S_{yy}} = \frac{4}{32 \times 18} = \frac{1}{144}$$

$$F = (n-2) \frac{r^2}{1-r^2} = 48 \times \frac{1/144}{143/144} = \frac{48}{143}$$